

Solution Requirements

Date	04 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

A comprehensive list of requirements defining exactly what the Credit Card Approval Prediction solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of the system, rather than specific behaviors (How the system should *be*).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Credit analysts and admins log in via a Flask-based login form with hashed password authentication; session-based auth restricts access to the prediction dashboard.	High
2	Authorization levels	Three roles: Applicant (self-service eligibility check only), Credit Analyst (submit and review predictions), Admin (manage models and view audit logs).	High
3	External interfaces	IBM Watson Machine Learning REST API for real-time model scoring; optional integration with a credit bureau API for credit history lookup.	High

4	Transactions processing	Each submitted application is validated, transformed into the model's feature format, scored by the deployed classifier, and the resulting prediction with confidence score is logged to the datastore.	High
5	Reporting	Dashboard showing approval/rejection counts, per-model accuracy metrics (Logistic Regression, Random Forest, XGBoost, Decision Tree), and an exportable CSV of processed applications.	Medium
6	Business rules, etc.	Applicants with a severe past-due payment history are auto-flagged high-risk regardless of model score; all required fields must be complete before scoring is allowed.	Medium
7	Compliance to laws or regulations	Applicant financial and identity data is handled per data-protection best practices; only fields necessary for scoring are retained; every decision is logged for auditability.	High
8	Other	The underlying model can be retrained or swapped (e.g., new XGBoost version) without changing the Flask API contract.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Time taken to return a prediction after an applicant submits their details.	Prediction returned in < 2 seconds per request

2	Scalability	The system must handle concurrent applicants during peak application periods.	Supports up to 500 concurrent scoring requests via IBM Watson ML auto-scaling
3	Security & Data Privacy	Applicant financial and identity data must be protected in transit and at rest.	TLS 1.2+ in transit, AES-256 encryption at rest
4	Reliability & Availability	The prediction service must remain available for continuous approval processing.	99.5% uptime per month
5	Usability & Accessibility	The web form must be simple enough for both analysts and applicants to use unaided.	Complies with WCAG 2.1 AA standards
6	Other	Maintainability: model versioning and a straightforward retraining pipeline.	New model version deployable within 1 business day